

EXB-04 · EXAM BLUEPRINT

How items are written, reviewed and retired

The item lifecycle the Institute is standing up — commissioning, authoring, review, bias review, pretesting, statistics and retirement

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

An examination is only as good as its item bank, and an item bank is a managed asset with a lifecycle rather than a folder of questions. This document sets out the lifecycle the Institute is standing up for the PCL-AI: how items are commissioned against the blueprint, the rules they are authored to, the three independent reviews they must survive, how they are pretested unscored, the statistics used to judge them once real candidates have answered them, and the triggers that retire them. It works the statistics through with illustrative figures and states plainly which parts of the process are running today and which are being built. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

— About this document

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How items are written, reviewed and retired

In one paragraph. An examination is only as good as its item bank, and an item bank is a managed asset with a lifecycle rather than a folder of questions. This document sets out the lifecycle the Institute is standing up for the PCL-AI: how items are commissioned against the blueprint, the rules they are authored to, the three independent reviews they must survive, how they are pre-tested unscored, the statistics used to judge them once real candidates have answered them, and the triggers that retire them. It works the statistics through with illustrative figures and states plainly which parts of the process are running today and which are being built. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

Who this is for. Subject-matter experts writing or reviewing items for the Institute; assessment professionals auditing our process; and practitioners who want to know why an examination body's quality claims should be believed.

— 1. The bank is the asset

The blueprint says what should be sampled. The item bank is what makes the sampling possible, and it is the single most expensive and most fragile thing a certification body owns.

Fragile, because an item bank degrades in four directions at once. Items leak, and a leaked item measures nothing. Practice changes, and an item that was correct in 2026 becomes wrong or merely quaint. The bank gets used, and over-exposed items stop discriminating. And every item carries a small probability of being defective in a way that no amount of expert authorship prevents and only candidate data reveals.

A lifecycle is the answer to all four. What follows is the process the Institute is building for the PCL-AI bank. §11 says honestly which parts of it are operating now.

— 2. The lifecycle in one view

Stage	Who	What happens	What stops an item here
1. Commission	Chief Examiner / Assessment Lead	A gap in the bank is identified against the blueprint and an item is commissioned to a specific knowledge area and cognitive level	—
2. Author	Subject-matter expert	The item is written to the authoring rules in §4, with a rationale for the key and for every distractor	—
3. Technical review	A second subject-matter expert	Correctness of the stem, the key, the arithmetic and every distractor rationale	Wrong key; unverifiable arithmetic; two defensible answers
4. Editorial review	Editorial	Clarity, format compliance, option balance, language, translation readiness	Cueing; excluded item types; protected text
5. Bias and fairness review	A panel independent of the author	Construct-irrelevant barriers: culture, jurisdiction, sector, language, accessibility	Anything that makes the item harder for reasons unrelated to competence
6. Pretest	—	The item is placed on live forms unscored , and candidate responses are collected	—
7. Statistical review	Assessment Lead with psychometric advice	Difficulty, discrimination and distractor performance are computed and read	Statistics outside the acceptance thresholds
8. Disposition	Chief Examiner	Accept into the scored bank, return for revision, or reject	—
9. In-bank management	Assessment Lead	Exposure tracking, periodic content review, version control	—
10. Retirement	Chief Examiner	The item leaves the scored bank on one of the triggers in §10	—

No item reaches a candidate's score without passing every stage. An item that fails a stage returns to its author with the reason; it is not quietly fixed by the reviewer, because a reviewer who edits an item has just reviewed their own work.

— 3. Commissioning: writing to a gap, not to a favourite topic

Items are commissioned against the blueprint, not offered against it. The Assessment Lead identifies where the bank is thin — a knowledge area with too few usable items, a cognitive level under-supplied in a particular group, a domain whose items are ageing — and commissions to that gap with the domain, knowledge area, cognitive level and, where relevant, the scenario setting specified in advance.

The alternative, which is how most banks accumulate, is to let experts submit what interests them. It produces a bank that is deep where the authors are strong and thin everywhere else, and it makes the blueprint aspirational: forms cannot be assembled to a specification the bank cannot supply.

Commissioning also spreads sector settings deliberately. A bank in which every scenario is a construction project examines construction professionals more gently than everyone else, and that is a fairness defect before it is an editorial one.

— 4. Authoring: the rules

Authors work to the format specification in [EXB-03 – Cognitive levels and item types](#) and to these rules.

One assessable point per item. If an item can be got wrong for two unrelated reasons, its statistics will be uninterpretable and its diagnostic value nil.

The stem stands alone. A competent candidate should be able to answer before seeing the options. Every figure needed is in the stem; nothing needed is outside the body of knowledge.

Four plausible options, of comparable length and parallel structure. The key is never systematically the longest, the most qualified or the most technically phrased. This is the most common defect in submitted items and the easiest for a candidate to exploit.

Every distractor names a real error. The author records, for each distractor, the specific mistake a practitioner would make to arrive at it — an inverted ratio, a cumulative figure used where an incremental one is needed, a method applied outside its assumptions, an action that contradicts the stem. A distractor without a named error is a filler option and is returned.

Arithmetic is shown in the rationale, not just the answer. $CPI = EV \div AC = 4,500,000 \div 5,000,000 = 0.90$. The reviewer recomputes from the substitution.

Currency-neutral, jurisdiction-neutral, vendor-neutral. Where a legal, tax or accounting treatment varies, the item states the frame as an assumption or avoids depending on it. Where a tool is involved, the item concerns the technique, never a product's interface.

Nothing protected is reproduced. Standards are described in the Institute's own words. No text, table, diagram or question from any external standard or examination is reproduced in any item.

No live content is ever published. Items written for publication — the sample items in [EXB-08 – Sample items with rationale](#) — are authored separately, marked as study material, and never enter the live bank. The separation runs in both directions: a live item is never published, and a published item is never scored.

— 5. Technical review: is it right?

A second subject-matter expert, who did not write the item, verifies four things.

The key is correct, and correct for the reason the author gives.

Every distractor is wrong, and wrong for the reason the author gives. This is where items most often fail. An author writes what they believe is a clearly inferior option; a reviewer from a different sector recognises it as standard practice in theirs. An item with two defensible answers is not a hard item, it is a broken one.

Every calculation is independently recomputed from the substitution, not checked against the author's answer. Numerical items get two technical reviewers.

The cognitive level tag matches what the item demands. An item tagged Analysis that resolves to a formula lookup is re-tagged or rewritten. Mis-tagged items corrupt form assembly silently, because the assembly rules trust the tag.

— 6. Bias and fairness review

This is a distinct review with a distinct question, and it is not a politeness check.

The question is: **does anything in this item make it harder for a group of candidates for reasons that have nothing to do with the competence being assessed?** Assessment calls that construct-irrelevant difficulty, and it is the mechanism by which a technically flawless item can still be unfair.

The panel reads every item for:

Jurisdictional assumption. An item that turns on one country's tax treatment, procurement law, retention practice or accounting rule tests where the candidate works, not what they know. The Institute's content is globally applicable; where treatment varies, the item states the frame as an assumption of the scenario.

Sector assumption. A scenario that requires knowledge of how a specific industry operates — commissioning sequences, shift patterns, regulatory gates — advantages that industry's practitioners. Scenarios carry the context they need inside the stem.

Cultural and idiomatic content. Idiom, humour, sporting or culinary metaphor, names and settings that read as familiar to some candidates and as noise to others. The examination interface runs in seven languages; an item that survives translation only as a paraphrase is a defective item.

Language complexity beyond the construct. Long subordinate clauses, rare vocabulary and unnecessary negation measure reading comprehension. Where a term is genuinely part of the discipline, it stays; where it is decoration, it goes.

Accessibility. No meaning carried by colour alone. No dependence on a visual layout that a screen reader cannot linearise. Figures and tables that read sensibly in sequence.

The panel is independent of the author, and it includes reviewers from more than one region and more than one delivery sector. A review panel drawn from a single background cannot see its own assumptions — which is the whole reason the stage exists.

Bias review before pretesting catches what can be reasoned about. It does not catch everything, which is why §7.3 exists.

— 7. Pretesting and item statistics

7.1 Why items are pretested unscored

A new item is placed on live forms as an **unscored** item. Candidates cannot tell which items are scored and which are being trialled, and pretest items do not contribute to anyone's result.

This is the only honest way to get the data. Item statistics computed on a volunteer sample of trainees describe trainees; statistics computed on real candidates under real examination conditions de-

scribe the population the item will actually be used on. The unscored placement means no candidate is ever penalised for an item the Institute has not yet validated.

An item is not scored until it has accumulated enough responses to make its statistics meaningful: [CONFIRM: minimum pretest response count before an item may be scored].

7.2 The two statistics that matter

Difficulty is the proportion of candidates who answered correctly — conventionally the p-value. Confusingly for newcomers, a high p-value means an easy item. On a four-option item, chance alone returns 0.25, so a p-value near or below 0.25 means candidates are not doing better than guessing.

Discrimination is whether the candidates who answered the item correctly were the candidates who were stronger overall. It is the more important of the two, because it is the property that makes an item useful: an item everyone gets right and an item everyone gets wrong both discriminate nothing, whatever their difficulty. The Institute computes a **discrimination index** — the proportion correct in the strongest group minus the proportion correct in the weakest group, using the conventional upper and lower 27 per cent of candidates by total score — alongside the item-total correlation.

A **negative discrimination index is the strongest signal in the whole process**. It means stronger candidates did worse on the item than weaker candidates, which almost always means the key is wrong, a distractor is also defensible, or the stem is misleading in a way that only expertise exposes.

7.3 Distractor analysis

The option counts are read as well as the totals. Three patterns get an item looked at:

- **A distractor drawing more responses than the key** — investigate immediately for a mis-key or a second defensible answer.
- **An option almost nobody selects** — a dead distractor. The item is effectively three-option, and its chance floor rises from 25 per cent to 33.3 per cent ($1 \div 3$), which changes what its difficulty means.
- **A distractor drawing strong candidates disproportionately** — usually an option that is correct under a reading the author did not anticipate.

Distractor analysis is also where bias review gets its second bite: an item whose statistics differ between candidate groups after controlling for overall ability is flagged for differential item functioning review. That analysis identifies that an item behaves differently; it never explains why, so a flagged item goes back to a human panel rather than being deleted on the strength of a number.

7.4 Worked example

Illustrative figures. Response counts are for a single pretest item; percentages are rounded to two decimal places and the discrimination index to two.

Item A — accepted. 200 candidates responded. Key is B.

Option	Responses	Share
A	24	12.0 %
B (key)	122	61.0 %
C	38	19.0 %
D	16	8.0 %

Difficulty: $p = 122 \div 200 = 0.61$ — moderately difficult, comfortably above the 0.25 chance floor.

Discrimination, using the upper and lower 27 per cent ($0.27 \times 200 = 54$ candidates in each group). Of the 54 strongest candidates, 45 answered correctly; of the 54 weakest, 14 did.

$$D = (45 \div 54) - (14 \div 54) = 0.8333 - 0.2593 = 0.57$$

A strongly discriminating item. Every distractor is alive — the smallest draws 8 per cent — so all three are doing work. **Disposition: accept into the scored bank.**

Item B — rejected. 200 candidates responded. Key is C.

Option	Responses	Share
A	61	30.5 %
B	58	29.0 %
C (key)	55	27.5 %
D	26	13.0 %

Difficulty: $p = 55 \div 200 = 0.275$ — barely above the 0.25 chance floor, and a distractor has out-drawn the key.

Discrimination: of the 54 strongest candidates, 16 answered correctly; of the 54 weakest, 18 did.

$$D = (16 \div 54) - (18 \div 54) = 0.2963 - 0.3333 = -0.04$$

Negative. Stronger candidates did slightly worse than weaker ones, and the responses are spread across three options almost evenly — the signature of candidates guessing between options they cannot separate. **Disposition: reject.** The item goes back with the finding that option A is very likely defensible under a reading the author did not intend; it is rewritten from the stem up, not patched.

The formal acceptance thresholds for difficulty and discrimination — the values at which an item is accepted, flagged for review or rejected — are [\[CONFIRM: item-acceptance thresholds for difficulty and discrimination\]](#), to be fixed with psychometric advice and published. The Institute would rather name that gap than publish a threshold chosen because it looks like the sort of number an examination body ought to have.

— 8. Disposition

Three outcomes, decided by the Chief Examiner on the evidence from stages 5 to 7:

Accept — the item enters the scored bank with its tags, its rationale and its statistics attached.

Revise — the item returns to its author with the specific finding. A revised item re-enters at stage 3 and is pretested again. Revision does not preserve the old statistics: an item whose stem or options have changed is a new item, and treating it otherwise is how contaminated data enters a bank.

Reject — the item is retained with its findings for the author's learning and never used. Rejected items are kept, not deleted, because a bank's rejection record is the evidence that its review actually bites.

— 9. In-bank management

An accepted item is not finished.

Exposure is tracked. Every use of an item on a form is recorded. An item used too often becomes predictable, and over-exposure is a security matter as much as a psychometric one — see [EXB-07 – Examination security and integrity](#).

Statistics are monitored continuously. An item's difficulty drifting downwards over time — more candidates getting it right without any change to the population — is one of the clearest indicators that the item has leaked.

Content is reviewed periodically. Every item carries a review date. Domain 13 items age fastest, because the practice they describe moves fastest; items resting on a standard that has been amended are reviewed on the amendment, not on the calendar.

Versions are controlled. Every item version carries an identifier, and every scored attempt records the bank version and answer-key version it was scored against. A result can therefore always be reconstructed exactly as it was computed, which is what makes an appeal answerable years later.

— 10. Retirement

An item leaves the scored bank on any of these triggers:

Trigger	Why
Suspected or confirmed exposure	A leaked item measures familiarity with the leak. Retirement is immediate and does not wait for proof.
Over-exposure	Use beyond the tracked threshold, whether or not a leak is suspected.
Statistical drift	Difficulty or discrimination moving outside the acceptance range on continuing use.
Content superseded	A standard amended, a practice changed, a technique obsolete. Common in Domain 13.
Blueprint change	A knowledge area removed or restructured in a major version of EXB-01 .
Fairness finding	A differential-item-functioning flag confirmed by panel review.

Retirement is permanent for that item version. Retired items are archived with their full history — statistics, review findings and retirement reason — because the archive is what allows a future re-

viewer to ask whether the bank's failures cluster anywhere, which is a question no live bank can answer about itself.

Results already issued are **not** disturbed by retirement. A candidate scored against the bank version in force at their sitting keeps that result; retiring an item changes future forms, not past ones. The single exception is a confirmed mis-key, where the affected results are rescored and every affected candidate is notified — including those whose result improves and those for whom nothing changes.

— 11. What is running today, and what is being built

The Institute is founding-stage, and this section says which is which rather than describing an aspiration in the present tense.

Operating now. Commissioning against the published blueprint. Authoring rules and the rationale requirement. Technical review by a second subject-matter expert with independent recomputation of every calculation. Editorial review. Item versioning, with the bank version and answer-key version recorded against every scored attempt, so any result can be reconstructed exactly.

Being stood up. The independent bias and fairness panel as a standing body with defined regional and sector composition. Systematic unscored pretesting on live forms. Routine computation and review of item statistics, and the acceptance thresholds those reviews will apply. Differential-item-functioning analysis, which needs candidate volumes the Institute does not yet have.

Dependent on volume. Several of the statistics above are meaningless below a certain number of responses, and no process design can create that data. Until the volumes exist, the Institute relies more heavily on the human review stages and says so, rather than reporting statistics that a small sample cannot support.

— 12. How this goes wrong

Experts assuming expertise is enough. The commonest failure in professional certification. A distinguished practitioner writes an item that is correct, important and unanswerable — or answerable from the options alone. Authorship and item writing are different skills, and the second is learned.

Reviewing your own item. A reviewer who edits rather than returns has removed the independence the stage exists to provide. The rule is absolute: findings go back to the author.

Patching a failed item instead of rewriting it. Changing one distractor on an item with negative discrimination usually leaves the underlying ambiguity in place, and the patched item carries the old statistics as though they still applied.

Treating difficulty as quality. A hard item that everyone answers by guessing is worse than an easy item that separates strong candidates from weak ones. Discrimination is the property that matters.

Letting the bank age quietly. No trigger fires when an item becomes out of date; someone has to look. The review date on every item is the mechanism, and it only works if it is enforced.

Publishing sample items drawn from the live bank. The single most damaging thing a certification body can do to its own examination, and it is almost always done with good intentions — a training partner asks for realistic practice material. The separation in §4 is not negotiable.

— 13. Checklist

For an item author, before submission:

- [] Domain, knowledge area and cognitive level are tagged, and the level matches what the item demands.
- [] The item can be answered from the stem alone.
- [] Options are of comparable length and parallel structure; the key is not the longest.
- [] Every distractor has a named practitioner error recorded in the rationale.
- [] Every calculation shows the substitution, and I have recomputed it from scratch.
- [] The item is jurisdiction-neutral, sector-neutral and vendor-neutral, or states its frame as an assumption.
- [] Nothing protected is reproduced.
- [] None of the excluded item types in [EXB-03 §7](#) is used.

For a reviewer:

- [] I did not write this item.
 - [] I attempted it with the options covered before reading them.
 - [] I recomputed every figure independently rather than checking the author's working.
 - [] I tested each distractor for whether it is defensible in a sector or jurisdiction other than my own.
 - [] My findings go back to the author; I have not edited the item myself.
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The reason to publish a lifecycle rather than assert quality is that a lifecycle can be audited and an assertion cannot. Anyone can say their items are written by experts. The question worth asking of any certification body is what happens to an item that the data says is broken — and whether the answer is a documented rejection with a retained record, or a quiet edit that nobody outside will ever see.

— Related

- [EXB-03 – Cognitive levels and item types](#) — the format specification these rules implement
 - [EXB-05 – Standard setting: modified Angoff and the pass mark](#) — where item difficulty enters the passing standard
 - [EXB-06 – Job-task analysis: where the blueprint gets its authority](#) — what tells the Assessment Lead where the bank should be deep
 - [EXB-07 – Examination security and integrity](#) — exposure, leakage and the protection of the bank
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— Sources and standards

The statistics in §7 are standard classical test theory measures — the proportion correct, the upper/lower discrimination index and item-total correlation — described here in the Institute's own words. The worked figures in §7.4 are illustrative and were independently recomputed before submission. The certification framework is developed with reference to ISO/IEC 17024 personnel-certification principles; PCI is not accredited by any ISO/IEC 17024 accreditation body.

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